Recipe 4

Short Read Aligners

A new era of biology

Alignments as Measurements

Paragdim

1. Correlate a biological phenomena with DNA
2. Measure that DNA

The measurements may be over hundreds of millions of sequence fragments.

Needs to be done efficiently.

Short Read Aligners

Marvels of algorithm development and implementation skills!

So hard to do that are usually implemented by just one person!

One guy, Heng Li, alone is responsible for the code `bwa` (Burrows-Wheeler Aligner that the generates all data for the 1000 genomes project and most clinical applications...

There is no other branch of science so dependent on just one person - think about that...

Tuned for a purpose

One person writes the code - it works the way they need it to work.

Which may be different of what you need it.

Alignments are more than just an arrangement - the additional information on the alignments matter a lot.

Best Aligner: Red Herring! 🐟

Much ink is spent on comparing aligners - most of it is a waste of time.

People fret about tiny changes in so called "accuracy" or "precision" or "recall" etc.

The "hidden" properties of the data have huge effect on the performance of each algorihtm.

See the book chapter: **How to compare aligners**

Decisions

Make the decision based on the following:

1. Can the aligner handle the volume of data relative to computational power?
2. Can the aligner be customizes to report the type of data you need? (all alignments or just best alignment, can you filter the output)
3. Will the aligner produce attributes of the alignment that you need? (alternative alignent locations, secondary alignments etc)

Recipe Code

The recipe runs two different short read aligners

1. Download the reference genome
2. Download the SRR sequencing data
3. Index the genome to prepare it for the algorithm
4. Perform the alignment and get a SAM file

Future recipes will explore the SAM format.